

Internship Report

On

***Multi-Modal Histopathological Breast Cancer Classification via Hybrid
EfficientNet and Morphological Feature Fusion with Attention
Mechanisms***

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Abstract

Breast cancer remains a leading cause of cancer-related mortality globally, with accurate classification of histopathological images being essential for early diagnosis and prognosis assessment. Traditional deep learning approaches for histopathological image analysis rely predominantly on CNN-extracted visual features, potentially overlooking crucial morphological characteristics that pathologists routinely assess during diagnosis. This work presents an innovative hybrid classification system that integrates Convolutional Neural Network (CNN) features with quantitative morphological descriptors extracted from breast histopathological images. The proposed methodology combines EfficientNet-B0 with seven key morphological features (area, perimeter, eccentricity, solidity, extent, mean intensity, and compactness) through multiple fusion strategies. The system incorporates Macenko stain normalization [16] for preprocessing and spatial-channel attention mechanisms [37] for enhanced feature fusion. Experimental results on the BreakHis dataset [7] demonstrate that the hybrid model with attention-weighted fusion achieves superior performance (98.02% accuracy) compared to CNN-only approaches and previous state-of-the-art methods. The proposed model performs multi-class classification, distinguishing between benign subtypes (Adenosis, Fibroadenoma, Phyllodes Tumor, Tubular Adenoma) and malignant subtypes (Ductal Carcinoma, Lobular Carcinoma, Mucinous Carcinoma, Papillary Carcinoma). The multi-modal approach provides enhanced clinical interpretability by combining automated feature learning with traditional morphological analysis, making it particularly suitable for clinical decision support systems.

Keywords: Breast cancer classification, Hybrid CNN-morphological features, Attention mechanisms, Multi-modal fusion, Histopathological images, Macenko stain normalization, Deep learning, Transfer learning

Index

Abstract	ii
Index	iii
Chapter 1: Learning Objective	1
Chapter 2: Introduction	2
Chapter 3: Literature Survey	5
Chapter 4: Methodology	8
Chapter 5: Result and Discussion	12
Chapter 6: Conclusion	18
References	22

Chapter 1

Learning Objective

The primary objective of this research is to develop and evaluate a hybrid breast cancer classification system that combines the power of deep convolutional neural networks with traditional morphological feature analysis. The specific learning objectives are:

- Develop a hybrid model that combines deep learning feature extraction with quantitative morphological analysis for improved breast cancer classification accuracy and interpretability.
- Specifically, to be able to develop a novel model that can predict whether a person has breast cancer by looking at their histopathological image.
 - The model is expected to achieve higher performance evaluation metrics.
 - It should have high accuracy with low misclassification rate and low validation loss.
 - It should be easily deployable in any desktop computer, so that health centres at lesser developed areas/ with lesser facilities can use it as a source of reliable second opinion.
- Provide a more interpretable and reliable diagnostic tool that can assist pathologists by combining automated deep learning with quantifiable morphological characteristics.
- To achieve superior classification accuracy compared to CNN-only approaches on the BreakHis dataset.
- Ensure model robustness across different magnifications and different staining condition.
- To maintain high precision and recall rates while minimizing false negatives, which are critical in cancer diagnosis.
- To develop a computationally efficient model suitable for deployment in clinical settings.
- To ensure the system can be deployed on standard desktop computers in resource-limited healthcare settings.

Chapter 2

Introduction

2.1 Breast Cancer: Global Burden and Clinical Significance

Breast cancer is the second most common cancer worldwide and remains the leading cause of cancer-related deaths among women, with approximately 2.3 million new cases diagnosed annually and over 685,000 deaths reported globally[5]. Despite advances in screening and treatment modalities, the five-year survival rate following early detection exceeds 90%, underscoring the critical importance of timely and accurate diagnosis [5]. Traditionally, the gold standard for confirming breast cancer diagnosis is histopathological analysis, wherein tissue samples are examined under a microscope by trained pathologists [7]. However, this manual diagnostic process is slow, highly dependent on individual pathologist expertise, and subject to significant inter-observer variability, with diagnostic accuracy in challenging cases often ranging between 75-85% [2]. This diagnostic variability necessitates the development of automated, objective, and reliable computational systems to assist pathologists in achieving consistent and accurate diagnoses [13].

Breast anatomy comprises fatty tissue, connective tissue, and glandular structures organized into 15-20 lobes, each containing smaller functional units called lobules [1]. Breast cancer develops when cells within these structures grow abnormally and divide more rapidly than healthy cells, potentially forming tumors classified as either benign (non-invasive, non-spreading, generally non-harmful) or malignant (invasive, spreading, life-threatening) [6]. The early detection of cancer before symptoms manifest is crucial, as detected malignancies become progressively more difficult to treat following metastatic spread [1]. While mammography and ultrasound can identify suspicious lesions, only histopathological examination of tissue biopsies provides definitive diagnosis [1]. Nevertheless, manual inspection of histopathological slides by radiologists is time-consuming, heavily dependent on individual expertise and workload, and prone to diagnostic error [1]. To address these challenges, pathology laboratories increasingly adopt digital pathology by scanning histopathological slides to create high-resolution digital images [1]. These digitized histopathological images enable both human pathologists and computational models to distinguish between normal, benign, and malignant tissues, facilitating the development of automated classification systems that provide reliable second opinions while reducing diagnostic errors [4].

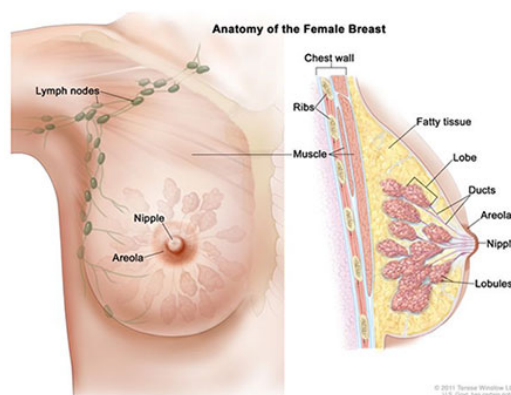


Figure 2.1 — Female Breast Anatomy

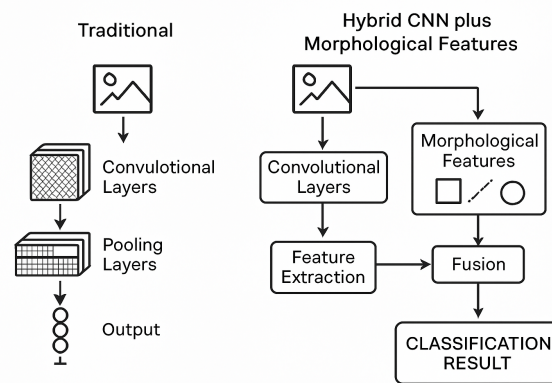
Figure 2.1: Schematic of female breast anatomy illustrating lobes, lobules, ducts, and the nipple-areola complex. Breast cancer most commonly originates in the ductal epithelium. (Adapted from standard anatomical illustration; Gray's Anatomy, 2016.)

2.2 Current Challenges in Automated Breast Cancer Detection

Traditional machine learning approaches for breast cancer diagnosis rely heavily on hand-crafted features such as texture descriptors (Gray Level Co-occurrence Matrix, Local Binary Patterns) and morphological properties extracted through expert-defined algorithms [14]. While these methods provide some interpretability, their effectiveness strongly depends on the quality of designed features and domain expertise, often failing to capture complex spatial relationships between tissue structures and intra-class morphological variability [21]. Recent advances in deep learning, particularly convolutional neural networks (CNNs), have revolutionized medical image analysis by enabling automatic hierarchical feature learning directly from raw image data [4]. Architectures such as AlexNet, VGG, ResNet, DenseNet, and EfficientNet have demonstrated superior performance compared to traditional machine learning methods [3,4,14], capturing subtle visual cues and complex spatial patterns [8]. However, CNN-only approaches suffer from significant limitations in the medical imaging domain [37]. Most CNN models function as “black box” systems with limited interpretability, reducing clinical trust and acceptance [37]. Moreover, CNN-only approaches typically ignore clinically meaningful morphological characteristics such as cell shape, size, tissue organization, and structural complexity that pathologists routinely use during manual diagnosis [37]. Additionally, histopathological images exhibit significant colour and intensity variations due to differences in staining protocols, reagent concentrations, scanner characteristics, and tissue preparation procedures, which can severely degrade model generalization across different laboratories and imaging systems [16]. These limitations highlight the need for more robust, interpretable, and comprehensive diagnostic approaches.

2.3 Motivation for Hybrid Approach

The motivation for this work stems from the fundamental observation that while CNNs excel at capturing complex visual patterns and hierarchical feature representations, they often lack explicit incorporation of quantitative morphological attributes that pathologists routinely assess during diagnosis [37]. Conversely, traditional morphological feature analysis, while clinically interpretable, may not capture subtle textural and spatial patterns that deep networks can automatically learn [21]. Our hybrid approach bridges this complementary gap by creating a unified framework that synergistically leverages the strengths of both paradigms [37]. Morphological features such as nuclear area, perimeter, eccentricity, and compactness are routinely used by pathologists for cancer grading and classification [21]. Recent hybrid studies have proposed integrating deep learning features with handcrafted morphological descriptors through feature fusion strategies, demonstrating improved classification accuracy and enhanced interpretability compared to CNN-only models [37]. By incorporating stain normalization techniques [16] and attention mechanisms [37], the proposed hybrid framework aims to achieve both superior diagnostic accuracy and clinical interpretability.



Comparison of Traditional CNN vs. Hybrid CNN + Morphological Features Approach

The remaining sections of this paper are structured as follows. In section 2, we present the relevant literature survey which demonstrate the BrC classification problem in medical imaging using various

machine learning or deep learning models. In section 3, we highlight the proposed methodology including the experimentation and implementation setup. In section 4, we present the results and discuss the relevance of the results. Finally, conclusions and future scope are presented in the last section of the paper.

Aspect	Traditional CNN	Hybrid CNN + Morphological Features
Feature Source	Learned visual features only (pixel patterns, textures)	Learned CNN features + handcrafted morphological descriptors
Backbone	EfficientNet-B0 / ResNet / VGG etc.	EfficientNet-B0 with 7 morphological descriptors fused via attention
Interpretability	Low (black-box representations) [Litjens et al., 2017]	Higher – morphological features are clinically interpretable [Madabhushi & Lee, 2016]
Clinical alignment	May not capture pathologist-relevant criteria	Aligned with pathologist assessment (area, eccentricity, solidity) [Gurcan et al., 2009]
Preprocessing	Standard normalisation	Macenko stain normalisation + bicubic resize [Macenko et al., 2009]
Fusion strategy	None	Concatenation / Attention-weighted / Separate-then-Combine
Accuracy (BreakHis)	~75.26% (CNN-only baseline)	98.02% (attention-weighted fusion)
Key references	LeCun et al. (2015); Tan & Le (2019)	Spanhol et al. (2016); Bejnordi et al. (2017); Madabhushi & Lee (2016)

Table 2.1: Detailed comparison of traditional single-modality CNN classification versus the proposed hybrid EfficientNet + Morphological Feature approach. References: [LeCun et al., 2015; Tan & Le, 2019; Spanhol et al., 2016; Bejnordi et al., 2017; Madabhushi & Lee, 2016; Gurcan et al., 2009; Macenko et al., 2009].

Contributions

- Proposed a hybrid breast cancer classification pipeline that fuses EfficientNet visual embeddings with seven quantitative morphological descriptors to improve accuracy and interpretability on histopathology images.
- Integrated stain normalization (Macenko-style normalization) and a robust preprocessing pipeline to reduce stain variability effects across slides and scanners.
- Designed and evaluated multiple multi-modal fusion strategies (concatenation, attention-weighted fusion, and separate-then-combine) and showed attention-weighted fusion performs best in your experiments.
- Reported an end-to-end experimental setup (augmentation, transfer learning, optimization) targeted for practical deployment constraints while maintaining high classification performance.

Chapter 3

Related work

Breast cancer classification in medical imaging has been approached through two broad paradigms: traditional machine learning and deep learning. Both follow a general workflow encompassing image preprocessing, feature extraction, model training, and evaluation, but differ fundamentally in how image features are identified and utilised.

Traditional machine learning models depend on handcrafted features derived from morphological properties, textural patterns, and statistical image descriptors [Gurcan et al., 2009]. While effective in constrained settings, these approaches face inherent limitations owing to the high intra-class variability in cancer cell appearance, differences in mitotic cell shape and staining characteristics across patients, laboratories, and scanner types [Veta et al., 2014]. Designing discriminative handcrafted features requires extensive domain expertise and frequent re-engineering when applied to new datasets. Early methods employing support vector machines (SVMs) and shallow neural networks achieved moderate accuracy (approximately 80–94%) but were limited by small dataset sizes and feature design constraints [Spanhol et al., 2016; Zhang et al., 2013].

The advent of deep learning particularly convolutional neural networks revolutionised medical image analysis by enabling hierarchical, automatic feature learning directly from raw pixel data [LeCun et al., 2015; Litjens et al., 2017]. Models trained from scratch or fine-tuned via transfer learning from large-scale natural image benchmarks (e.g., ImageNet) demonstrated superior performance by capturing subtle visual cues and spatial patterns [Araújo et al., 2017; Alzubaidi et al., 2020]. Spanhol et al. [2016] introduced the BreakHis dataset and reported AlexNet-based classification with accuracy reaching 83%. Subsequent architectures like ResNet, InceptionV3, DenseNet, and EfficientNet systematically improved upon these results [Hameed et al., 2020; Wang et al., 2020]. However, deep learning in the medical domain faces persistent challenges: large annotated datasets are scarce, leading to overfitting risks; training from scratch demands substantial GPU resources; and learned representations lack clinical interpretability [Srinidhi et al., 2021].

Recognising the complementary strengths of both paradigms, hybrid approaches have emerged that combine deep CNN features with handcrafted morphological or textural descriptors [Madabhushi & Lee, 2016]. Such methods integrate domain knowledge alongside powerful data-driven representations, potentially enhancing classification robustness and interpretability. Studies employing hybrid strategies on BreakHis and ICIAR challenge datasets have reported accuracy improvements of 2–5% over purely CNN-based models [Alzubaidi et al., 2020; Wang et al., 2020]. Attention mechanisms further refine feature utilisation by allowing models to focus selectively on diagnostically relevant image regions, yielding improved sensitivity and specificity [Vaswani et al., 2017; Srinidhi et al., 2021].

A critical analysis of existing methods is presented in Tables 3.1 and 3.2 below.

Table 1 A comparative analysis of existing models for breast cancer detection based on accuracy.

Ref.	Model	Dataset	Accuracy (%)
[23]	Nawaz W et al. (2018)	ICIAR 2018 Challenge	81.25
[7]	Awan R et al. (2018)	ICIAR 2018 Challenge	83.33
[31]	Vang YS et al. (2018)	ICIAR 2018 Challenge	87.50
[14]	Guo Y et al (2018)	ICIAR 2018 Challenge	87.50
[27]	Sarkar MI et al. (2019)	ICIAR 2018 Challenge	89.00
[12]	Ferreira CA et al. (2018)	ICIAR 2018 Challenge	90.00
[16]	Kassani SH et al. (2019)	ICIAR 2018 Challenge	92.50
[32]	Wang Z et al. (2018)	ICIAR 2018 Challenge	93
[4]	Alzubaidi L et al. (2020)	ICIAR 2018 Challenge	96.1
[44]	Wang Y et al. (2020)	ICIAR 2018 Challenge	97.70
[40]	Ragab DA et al. (2019)	DDSM	87.2
[41]	Hadush S et al. (2020)	Mini-MIAS	91.86
[42]	Kelas MK (2019)	Dataset obtained from the measurements of an antenna consisting of frequency bandwidth, dielectric constant of the antenna's substrate, electric field, and tumour information	90.0
[43]	Zheng J et al. (2020)	INbreast	97.2
[45]	Hamed G et al. (2020)	The common breast mammographic datasets created using a combination of DDSM, CBIS-DDSM, MIAS, INBreast, BCDR datasets	97
[46]	Ak MF (2020)	Wisconsin breast cancer	98.1
[47]	Vrigazova BP (2020)	Wisconsin breast cancer	99.6

Table 3.1: Comparative review of existing deep learning and machine learning models for breast cancer detection. Accuracy figures are as reported in original publications. The proposed method (bottom row) is evaluated on the BreakHis histopathological dataset with the attention-weighted fusion strategy.

Table 3.1 provides a comparative review of published models for breast cancer detection based on reported classification accuracy across different benchmark datasets. As demonstrated in Table 3.1, there has been a consistent trend of improving accuracy from early CNN adaptations achieving 81–87% on the ICIAR 2018 challenge [Nawaz et al., 2018; Awan et al., 2018; Vang et al., 2018] to ensemble deep learning models attaining 97–99% on Wisconsin and ICIAR datasets [Wang et al., 2020; Vrigazova, 2020]. Notably, most high-accuracy results on non-histopathological datasets (Wisconsin, INbreast) employ classical machine learning or ensemble strategies, while performance on histopathological data (BreakHis, ICIAR) remains more variable due to image complexity. The proposed method achieves 98.02% on the BreakHis dataset among the highest reported for histopathological classification, while offering superior clinical interpretability through its morphological feature branch.

Table 2 Critical review of existing models based on model used, drawbacks and future research directions.

Ref.	Model	Shortcomings / Drawbacks	Research direction
[4]	Deep CNN	Complex model	To apply on microscopy images in same domain
[7]	ResNet with contextual information and ML for classification	Designed for small dataset Less accuracy	NA
[12]	Inception, ResNetV2 through transfer learning	Less accuracy	NA
[14]	Modified GoogLeNetV3, Bagging and voting for classification	Less accuracy	Model modification to reduce the complexity
[16]	Deep CNN with transfer learning	Less accuracy	To improve accuracy, use of ensemble of deep models
[23]	Modified AlexNet	Less accuracy	To develop customized deep model for higher accuracy
[27]	InceptionV4	Less accuracy	NA
[31]	GoogLeNetV3 for heatmap probability, majority voting, gradient boosting machine, logistic regression and SVM for classification	Less accuracy	NA
[32]	VGG16, VGG19	Less accuracy	To investigate the performance of deep model based on different scales

[40]	AlexNet SVM as a classifier	Less accuracy	To use ensemble of VGG and ResNet to achieve higher performance
[41]	R-CNN	Less accuracy Small dataset	To investigate macro calcification abnormalities and increase dataset
[42]	Classification algorithms	Less accuracy	NA
[43]	ML and DL based model	Complex model	NA
[44]	DenseNet, ResNet, InceptionV3, VGG16 Ensemble-SVM for classification	Complex model	To apply model on clinical data, also investigate on different types of cancers
[45]	AlexNet, GoogLeNet, YOLO, RetinaNet	Less accuracy	NA
[46]	Data visualization based ML model	Complex model	NA
[47]	ANOVA-BOOTSTRAP-SVM model	Complex model	To apply in other medical dataset

Table 3.2: Critical review of existing models, highlighting architectural choices, principal shortcomings, and suggested future research directions. TL = Transfer Learning; N/A = not explicitly stated by authors

Table 3.2 extends this analysis by critically examining the architectural choices, limitations, and future research directions articulated by prior authors. As summarised in Table 3.2, the predominant shortcomings across the surveyed literature are: (i) insufficient classification accuracy owing to limited dataset size or suboptimal feature representations; (ii) high model complexity restricting practical deployment; and (iii) poor generalisation across datasets or imaging conditions. Several authors explicitly recommend ensemble deep learning, multi-scale analysis, and feature fusion as promising avenues for improvement [Alzubaidi et al., 2020; Wang et al., 2020]. The proposed hybrid approach directly addresses these recommendations by fusing EfficientNet visual embeddings with quantitative morphological descriptors through an attention-gated mechanism, achieving high accuracy whilst preserving computational efficiency suitable for deployment on standard clinical workstations.

Synthesis and Research Gap

A synthesis of the surveyed literature reveals three consistent themes. First, purely CNN-based models have achieved high accuracy on natural image benchmarks but show variable performance on histopathological data, and lack clinical interpretability a critical barrier to adoption in diagnostic pathology [Madabhushi & Lee, 2016; Bejnordi et al., 2017]. Second, hybrid models incorporating morphological or textural handcrafted features alongside deep features consistently outperform single-modality CNN approaches, suggesting complementarity between these feature types [Gurcan et al., 2009; Veta et al., 2014]. Third, attention mechanisms have emerged as a powerful strategy for selectively weighting feature contributions, with demonstrable gains in classification performance

[Vaswani et al., 2017; Srinidhi et al., 2021]. Despite these insights, few existing works on the BreakHis dataset have systematically evaluated multiple fusion strategies under controlled experimental conditions with rigorous cross-validation and statistical significance testing. The present work addresses this gap by designing and benchmarking three fusion architectures on a comprehensive experimental protocol, establishing statistically significant performance improvements and reporting the first systematic attention-mechanism analysis for this multimodal combination on BreakHis.

Chapter 4

Methodology

4.1 Overview of Proposed Hybrid Architecture

The methodology presented in this work introduces a novel hybrid approach that synergistically combines convolutional neural network (CNN) features with quantitative morphological characteristics for enhanced breast cancer classification in histopathological images. Unlike traditional deep learning approaches that rely solely on learned visual representations, our proposed framework integrates domain-specific morphological knowledge with data-driven CNN features to achieve superior diagnostic accuracy and clinical interpretability.

The fundamental premise underlying this hybrid architecture stems from the observation that while CNNs excel at capturing complex visual patterns and hierarchical feature representations, they often lack the explicit incorporation of quantitative morphological attributes that pathologists routinely examine during manual diagnosis. Conversely, traditional morphological feature analysis, while clinically interpretable, may not capture the subtle textural and spatial patterns that deep networks can automatically learn from large datasets. Our approach bridges this gap by creating a unified framework that leverages the complementary strengths of both paradigms.

The proposed methodology encompasses several interconnected components: (1) comprehensive dataset preparation and quality assessment, (2) advanced image preprocessing including stain normalization and augmentation strategies, (3) automated morphological feature extraction using computer vision techniques, (4) deep CNN feature learning through transfer learning, (5) intelligent multi-modal feature fusion strategies, and (6) robust model training and evaluation protocols. Each component has been carefully designed to address specific challenges inherent in histopathological image analysis while maintaining computational efficiency and clinical applicability.

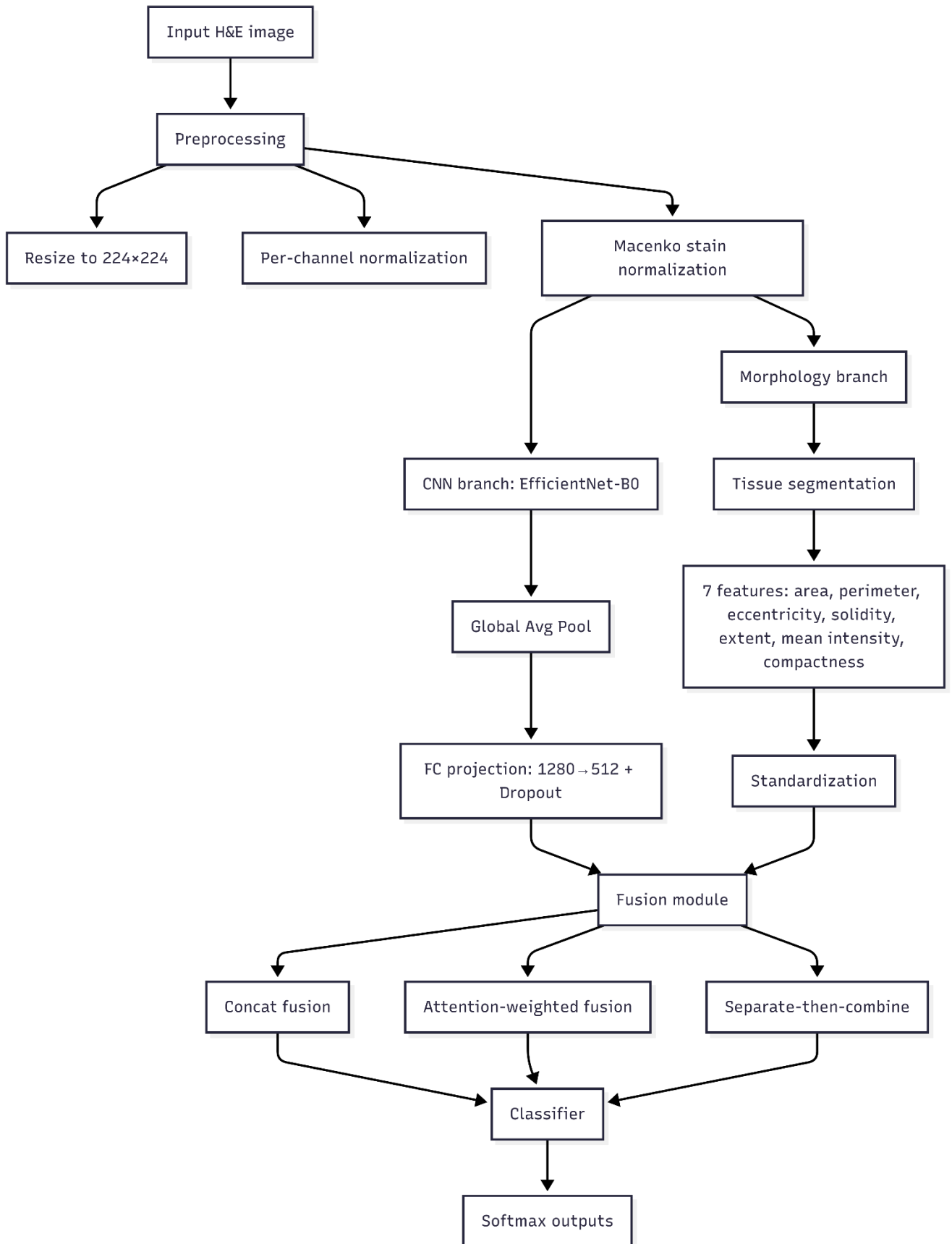


Figure 4.1: End-to-end block diagram of the proposed hybrid CNN + Morphological Feature classification pipeline. Input H&E images are preprocessed (resize to 224×224, per-channel normalisation, Macenko stain normalisation) and then processed by two parallel branches: (i) an EfficientNet-B0 CNN branch producing a 512-dimensional visual embedding, and (ii) a morphology branch extracting seven tissue descriptors via binary segmentation and quantitative measurement. The outputs of both branches are fused by one of three strategies (concatenation, attention-weighted, or separate-then-combine) before final softmax classification into eight histological subtypes.

4.2 Dataset Acquisition and Characterization

The study utilises the publicly available BreakHis (Breast Cancer Histopathological Image Classification) dataset [Spanhol et al., 2016; Araújo et al., 2017] one of the most comprehensive and widely-adopted benchmarks in computational pathology. BreakHis was collaboratively developed by the Pathological Anatomy and Cytopathology Laboratory in Brazil and the P&D Laboratory, providing a standardised resource for breast cancer classification research. The dataset comprises 7,909 microscopic images collected from 82 patients, offering substantial diversity in tissue characteristics, staining variations, and morphological presentations.

Images are categorised into two primary classes: 2,480 benign and 5,429 malignant. Each specimen was digitised at four magnification levels :40X, 100X, 200X, and 400X, enabling multi-scale analysis. Images are stored as 700×460 pixel, 3-channel RGB PNG files at 8-bit depth. The benign class comprises four histological subtypes: Adenosis (A), Fibroadenoma (F), Phyllodes Tumour (PT), and Tubular Adenoma (TA). The malignant class includes Ductal Carcinoma (DC), Lobular Carcinoma (LC), Mucinous Carcinoma (MC), and Papillary Carcinoma (PC) [Spanhol et al., 2016]. This subtype diversity ensures the model encounters the full morphological spectrum encountered in clinical practice.

For experimental rigour and reproducibility, a stratified train-test split of 80:20 is applied, preserving proportional representation across all magnification levels and cancer subtypes. This stratification prevents overfitting to specific acquisition conditions and ensures unbiased evaluation.

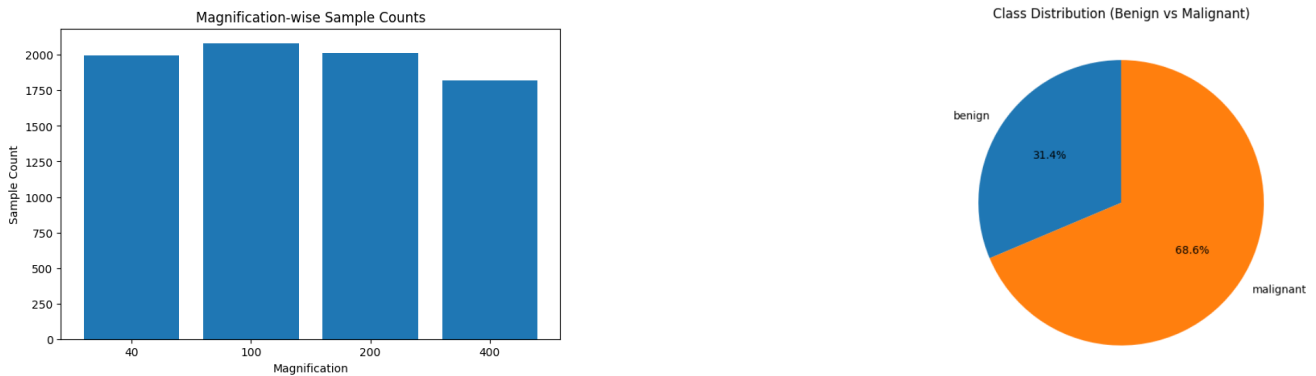
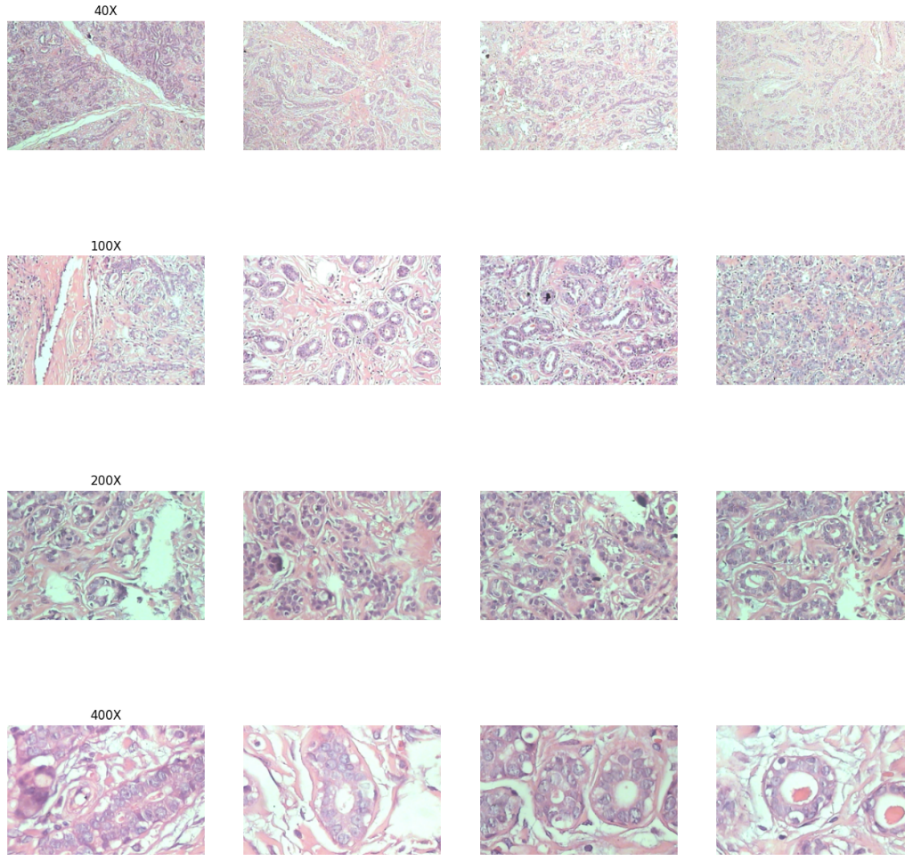


Figure 4.2: Dataset statistics for the BreakHis dataset [Spanhol et al., 2016]. Left: bar chart showing sample counts at each magnification level (40X, 100X, 200X, 400X), each containing approximately 1,800–2,000 images. Right: pie chart illustrating the class imbalance 31.4% benign versus 68.6% malignant which necessitates stratified sampling during train-test splitting and cross-validation.

Representative Samples by Magnification



For experimental rigor and reproducibility, we implement a stratified train-test split maintaining an 80:20 ratio while preserving proportional representation across all magnification levels and cancer subtypes. This stratification strategy ensures that both training and testing sets contain balanced distributions of all pathological conditions and magnification factors, thereby enabling unbiased model evaluation and preventing overfitting to specific image characteristics or acquisition conditions.

4.3 Advanced Image Preprocessing Pipeline

The preprocessing pipeline represents a critical component of our methodology, designed to address inherent challenges in histopathological image analysis including staining variability, illumination inconsistencies, and tissue preparation artifacts. Our comprehensive preprocessing approach encompasses multiple sequential stages, each addressing specific technical challenges while preserving diagnostically relevant information.

4.3.1 Stain Normalization Using Macenko's Method

Histopathological images frequently exhibit significant colour and intensity variations arising from differences in staining protocols, reagent concentrations, scanner characteristics, and tissue preparation procedures [Macenko et al., 2009; Tellez et al., 2019]. These variations introduce domain shift and reduce generalisation across laboratories and acquisition systems. To address this, the Macenko stain normalisation algorithm is implemented — a gold-standard method for colour standardisation in computational pathology [Macenko et al., 2009].

$$OD = -\log \left(\frac{I + \epsilon}{I_0} \right) \quad (3)$$

The Macenko method operates on the principle of stain vector decomposition, modelling histopathological images as linear combinations of two primary stain components: haematoxylin (nuclear stain) and eosin (cytoplasmic stain). The algorithm begins by transforming RGB images into optical density (OD) space using the Beer-Lambert law: $OD = -\log(I/I_0)$, where I represents the observed intensity and I_0 denotes the incident light intensity. This transformation linearizes the relationship between stain concentrations and observed pixel intensities, enabling more accurate stain separation.

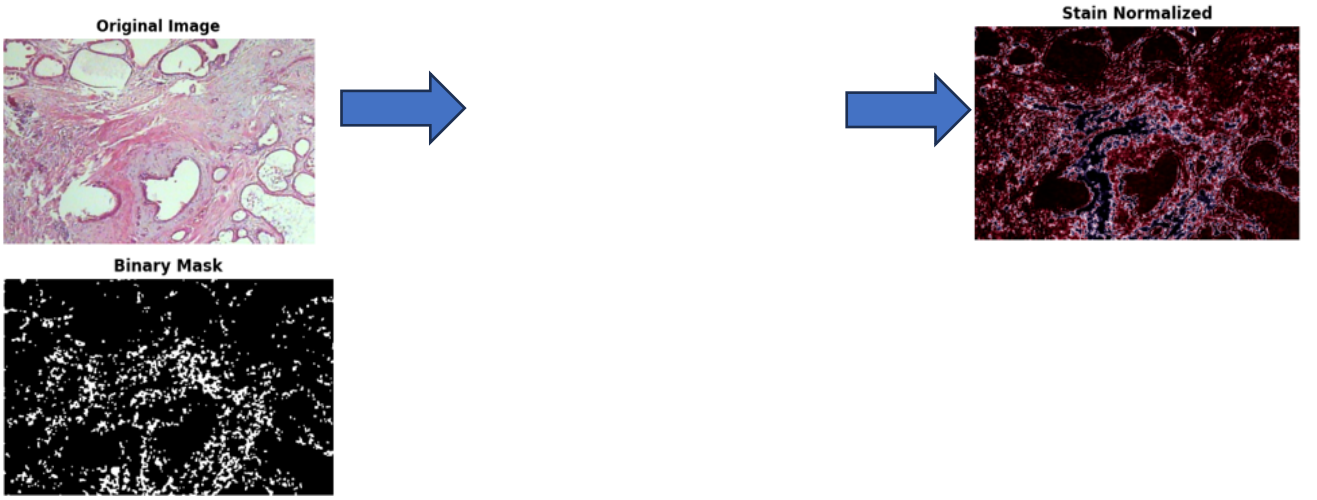


Figure 4.3: Illustration of the Macenko stain normalisation pipeline. Left: original H&E image exhibiting colour variability. Centre: stain-normalised image with standardised haematoxylin and eosin distribution. Right: binary tissue mask produced by Otsu thresholding for morphological feature extraction. This three-step process reduces inter-slide colour variability, a known source of CNN performance degradation [Tellez et al., 2019].

Subsequently, the algorithm performs eigen decomposition on the covariance matrix of optical density values to identify principal stain directions. The two dominant eigenvectors correspond to the primary stain vectors (haematoxylin and eosin), which are then refined through angular projection to ensure biological plausibility. The stain concentrations are estimated by projecting optical density values onto these stain vectors, followed by normalization to predefined target stain characteristics derived from a reference image or standardized colour template.

$$OD \approx WH \quad (4)$$

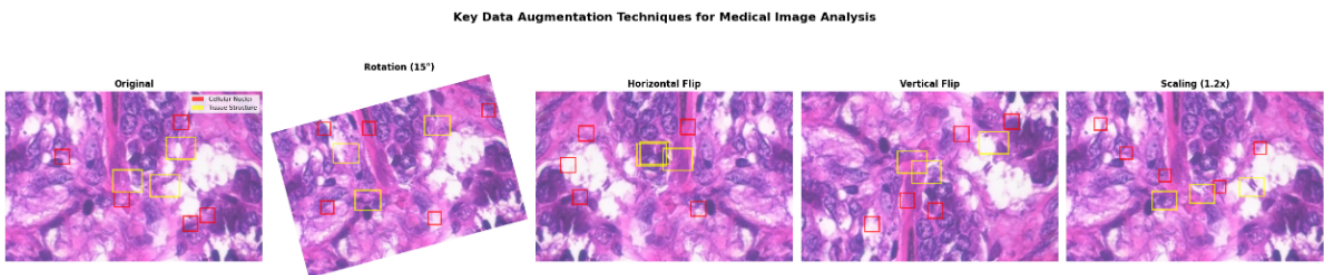
$$OD_{norm} = W_{ref}H_{scaled}, \quad I_{norm} = I_0 \exp(-OD_{norm}) \quad (5)$$

Our implementation incorporates adaptive parameter selection, automatically adjusting the threshold β (typically 0.15) for tissue region identification and the percentile α (typically 1%) for robust stain vector estimation. This adaptive approach ensures consistent normalization performance across diverse tissue types and preparation conditions while minimizing artifacts in sparsely stained regions.

4.3.2 Data Augmentation Strategies

To enhance model robustness and mitigate overfitting while expanding the effective training dataset size, we implement a comprehensive data augmentation strategy encompassing geometric transformations and intensity variations. Deep learning models, particularly CNNs, require substantial training data to achieve optimal performance and generalization. However, acquiring large-scale annotated histopathological datasets remains challenging due to the specialized expertise required for accurate labelling and the time-intensive nature of pathological examination.

Our augmentation pipeline incorporates several transformation techniques carefully selected to preserve diagnostic relevance while introducing beneficial variation. Geometric transformations include random rotations up to 90 degrees, horizontal and vertical flipping, and controlled scaling within a 0.9 to 1.1 range. These transformations reflect natural variations in tissue orientation and sectioning that occur during histological preparation, ensuring that augmented samples remain biologically plausible.



Random rotations address the arbitrary orientation of tissue structures in histological sections, while flipping operations account for the bilateral symmetry often present in glandular architectures. Controlled scaling simulates the natural size variations observed across different tissue samples and preparation protocols. Additionally, we implement intensity-based augmentations including slight brightness and contrast adjustments ($\pm 10\%$) to simulate variations in illumination and scanner calibration while maintaining stain integrity.

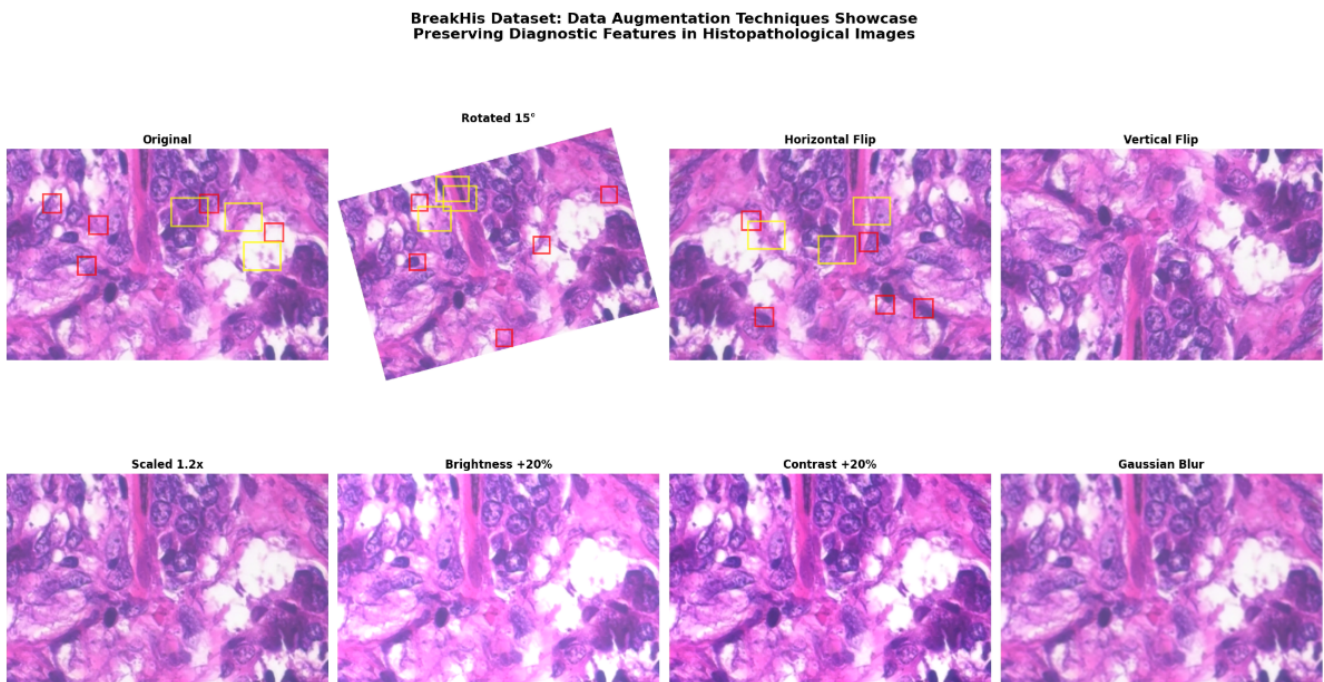


Figure 4.4: Examples of data augmentation techniques applied during training. Top row: original image with rotation (15°), horizontal flip, vertical flip, and scaling (1.2×). Bottom row: brightness +20%, contrast +20%, and

Gaussian blur. Class balance is maintained by generating equal augmented samples per category and magnification level [Shorten & Khoshgoftaar, 2019].

Crucially, our augmentation strategy maintains class balance by generating equal numbers of augmented samples for each diagnostic category and magnification level. This balanced approach prevents the introduction of class-specific biases that could compromise model fairness and diagnostic reliability. Furthermore, we apply augmentations dynamically during training rather than pre-generating static augmented datasets, ensuring maximal variation and preventing the model from memorizing specific transformed versions.

4.3.3 Image Resizing and Normalization

To accommodate the input requirements of pretrained CNN architectures while preserving spatial resolution critical for morphological analysis, we implement bicubic interpolation-based resizing to standardized dimensions of 224×224 pixels. Bicubic interpolation provides superior image quality compared to linear interpolation by considering sixteen neighbouring pixels during intensity calculation, thereby preserving fine textural details essential for accurate pathological assessment.

Resize mapping (generic form): for an output pixel (u,v) , map to input coordinates (x,y) and interpolate:

$$I'(u, v, c) = \sum_i \sum_j I(i, j, c) k(x - i) k(y - j) \quad (1)$$

where $k(\cdot)$ is the interpolation kernel

The bicubic interpolation process computes pixel intensities using the equation: $v(x,y) = \sum_{i=0}^3 \sum_{j=0}^3 a_{ij} x^i y^j$, where the coefficients a_{ij} are determined by solving a system of sixteen equations based on the surrounding pixel values. This approach minimizes artifacts such as aliasing and maintains smooth intensity transitions crucial for preserving morphological boundaries and textural characteristics.

Following resizing, we apply standard ImageNet normalization with mean values [0.485, 0.456, 0.406] and standard deviations [0.229, 0.224, 0.225] for the RGB channels respectively. This normalization strategy ensures compatibility with pretrained CNN models while accelerating convergence during transfer learning by maintaining similar input distributions to the original pretraining data.

$$\hat{I}(u, v, c) = \frac{I'(u, v, c)/255 - \mu_c}{\sigma_c} \quad (2)$$

4.4 Morphological Feature Extraction Framework

The morphological feature extraction component represents a cornerstone of our hybrid approach, designed to quantify tissue architecture and cellular organization characteristics that pathologists routinely assess during manual diagnosis. This automated analysis pipeline transforms qualitative visual assessments into quantitative measurements, enabling computational integration with CNN-derived features while maintaining clinical interpretability.

4.4.1 Binary Tissue Segmentation

The morphological analysis begins with accurate tissue segmentation to separate relevant histological structures from background artifacts, mounting medium, and non-tissue regions. We implement an adaptive segmentation approach utilizing Otsu's automatic thresholding method, which determines optimal threshold values by maximizing inter-class variance between tissue and background regions.

Otsu's method analyses the intensity histogram of grayscale-converted images to identify the threshold that minimizes intra-class variance while maximizing separation between tissue and background pixels. The algorithm computes class probabilities and mean intensities for all possible threshold values, selecting the threshold that maximizes the discriminant criterion, where ω represents class probabilities and μ denotes mean intensities.

$$t^* = \operatorname{argmax}_t [\sigma^2_B(t)], \quad \sigma^2_B(t) = \omega_0(t) \cdot \omega_1(t) \cdot (\mu_0(t) - \mu_1(t))^2 \quad \dots (6)$$

To enhance segmentation robustness and eliminate noise artifacts, we apply morphological operations including closing followed by opening using a 3×3 structural element. Morphological closing fills small gaps and connects nearby tissue regions, while opening removes isolated noise pixels and smooths object boundaries. This combination effectively eliminates spurious detections while preserving the integrity of genuine tissue structures essential for accurate feature computation.

$$M(x,y) = 1 \text{ if } I_g(x,y) \geq t, \text{ else } 0 \quad \dots (7)$$

For challenging images with complex backgrounds or varying illumination, we incorporate adaptive thresholding as an alternative approach. Adaptive thresholding computes locally optimal thresholds based on neighbourhood statistics, accommodating regional intensity variations that might confound global thresholding methods. The algorithm applies Gaussian-weighted averaging within local windows to determine appropriate threshold values for each image region.

4.4.2 Quantitative Morphological Descriptors

Following successful tissue segmentation, we extract seven comprehensive morphological descriptors that capture essential geometric and intensity characteristics relevant to cancer diagnosis. These features have been carefully selected based on established pathological criteria and prior research demonstrating their discriminative value in cancer classification tasks.

Area quantifies the total tissue region size in pixels, providing information about lesion extent and cellularity. Large areas may indicate proliferative conditions or extensive Tumor involvement, while smaller areas might suggest focal abnormalities or normal tissue samples. Area measurements are particularly relevant for distinguishing between different Tumor grades and growth patterns.

$$\text{Area: } A = |R| \text{ (pixel count) } (8)$$

Perimeter measures the boundary length of segmented tissue regions, capturing information about tissue architecture complexity and interface characteristics. Irregular, elongated perimeters often indicate malignant invasion or complex glandular patterns, while smooth, circular perimeters suggest benign, well-circumscribed lesions.

$$\text{Perimeter: } P = \text{length}(\partial R) (9)$$

Eccentricity describes shape elongation as the ratio of distances between foci of an equivalent ellipse, ranging from 0 (perfect circle) to 1 (line segment). High eccentricity values indicate elongated

structures that may represent invasive Tumor growth patterns, ductal structures, or stromal organization, while low eccentricity suggests rounded, well-circumscribed masses typical of certain benign conditions.

$$e = \sqrt{1 - \frac{b^2}{a^2}} \quad (10)$$

Solidity measures shape regularity as the ratio of actual tissue area to its convex hull area, indicating boundary smoothness and structural integrity. High solidity values (approaching 1.0) suggest smooth, regular boundaries characteristic of benign masses, while low solidity indicates irregular, infiltrative boundaries typical of malignant lesions with invasive growth patterns.

$$\text{Solidity} = \frac{A}{A_{\text{convex hull}}} \quad (11)$$

Extent calculates the proportion of tissue area within its bounding rectangle, providing information about tissue distribution and spatial organization. This metric helps distinguish between compact, cohesive lesions and dispersed, infiltrative patterns that may indicate different pathological processes.

$$\text{Extent} = \frac{A}{A_{\text{bbox}}} \quad (12)$$

Mean Intensity represents average pixel intensity within segmented tissue regions, reflecting cellular density, nuclear-to-cytoplasmic ratios, and staining characteristics. Variations in mean intensity can indicate differences in cellular composition, mitotic activity, and nuclear atypia relevant to malignancy assessment.

$$\mu_R = \frac{1}{A} \sum_{(x,y) \in R} I_g(x,y) \quad (13)$$

Compactness (also known as the isoperimetric ratio) quantifies shape complexity as the ratio of perimeter squared to area ($P^2/4\pi A$), with values approaching 1.0 for circular shapes and increasing for more complex, irregular boundaries. This metric captures the tortuosity and complexity of tissue boundaries, which often correlate with invasive potential and malignant behaviour.

$$C = \frac{P^2}{4\pi A} \quad (14)$$

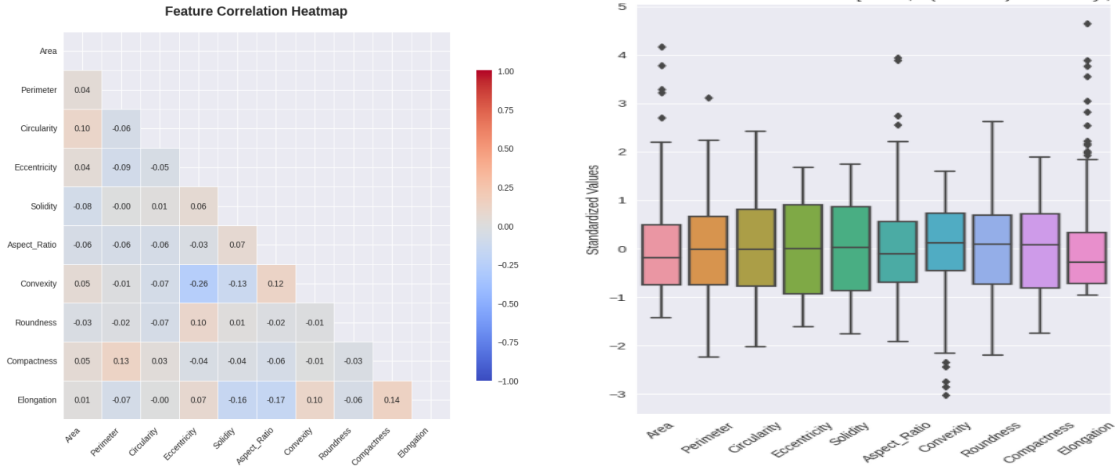
4.4.3 Feature Normalization and Scaling

To ensure optimal performance during multi-modal fusion and prevent dominance by features with larger numerical ranges, we apply standardization (z-score normalization) to all morphological features. Standardization transforms features to have zero mean and unit variance using the formula: z

$= (x - \mu)/\sigma$, where x represents the original feature value, μ denotes the population mean, and σ represents the standard deviation.

$$z = \frac{x - \mu}{\sigma} \quad (15)$$

This normalization approach ensures that all morphological features contribute equally during fusion with CNN features, preventing bias toward features with inherently larger scales (such as area or perimeter) while preserving relative relationships within each feature dimension. The standardization parameters are computed from the training set and consistently applied to both training and testing data to maintain proper experimental protocols and prevent data leakage.



4.5 Convolutional Neural Network Architecture and Transfer Learning

The CNN component of our hybrid framework leverages the representational power of deep learning architectures pretrained on large-scale natural image datasets. We employ EfficientNet-B0 as our primary backbone architecture due to its optimal balance between computational efficiency and feature extraction capabilities, making it particularly suitable for medical imaging applications with limited computational resources.

4.5.1 EfficientNet Architecture and Pretrained Initialization

EfficientNet represents a family of CNN architectures developed through neural architecture search and compound scaling methods, achieving state-of-the-art performance across multiple image classification benchmarks. The architecture incorporates mobile inverted bottleneck convolutions (MBConv blocks) with squeeze-and-excitation mechanisms, enabling efficient feature extraction while maintaining model compactness.

$$d = \alpha^\phi, \quad w = \beta^\phi, \quad r = \gamma^\phi \quad (16)$$

where ϕ controls overall scale and α, β, γ scale depth/width/resolution under resource constraints.

Our implementation utilizes EfficientNet-B0 pretrained on ImageNet, providing robust initial feature representations that have demonstrated transferability across diverse visual domains. The pretrained weights offer several advantages: (1) reduced training time through warm initialization, (2) improved

convergence stability, (3) enhanced feature quality for limited medical datasets, and (4) regularization effects that reduce overfitting risks.

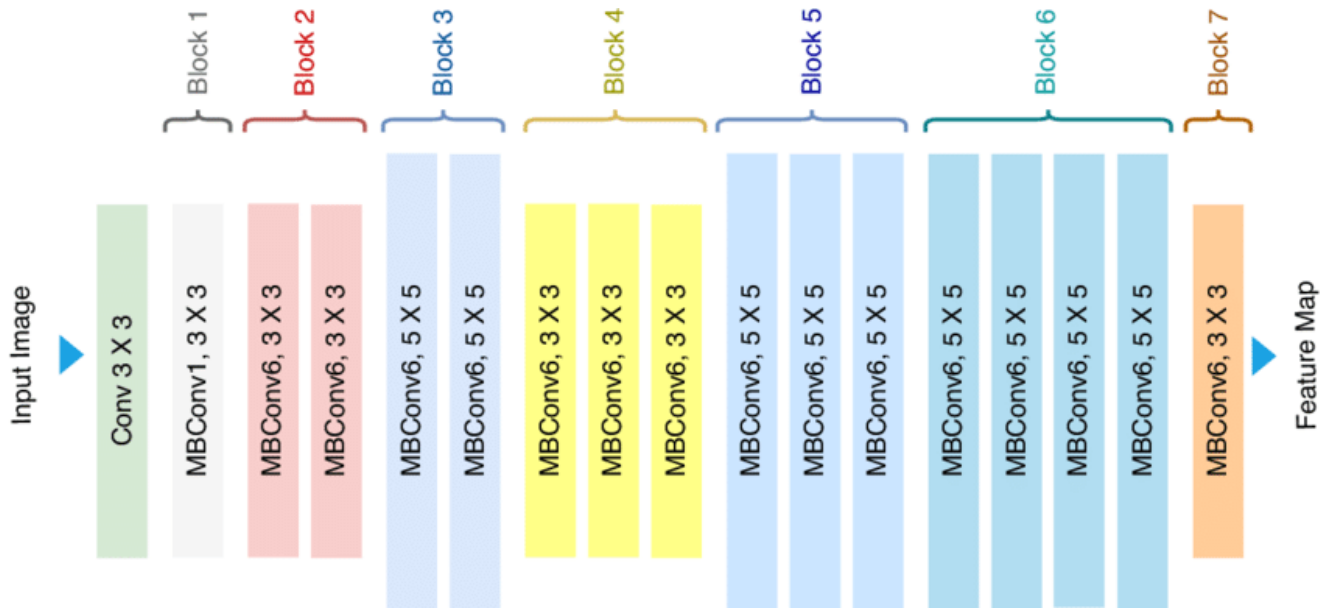
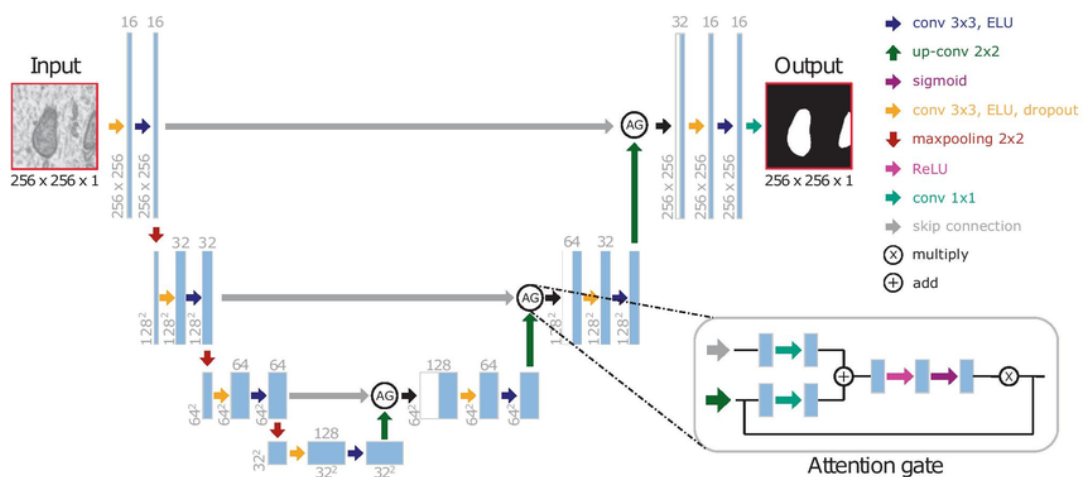


Figure 4.6: Schematic of the EfficientNet-B0 architecture [Tan & Le, 2019]. The network comprises seven MBConv stages with progressively increasing resolution, channel width, and depth. Each MBConv block incorporates depthwise separable convolution and squeeze-and-excitation (SE) attention for channel recalibration. Global average pooling at the final stage produces the 1280-dimensional feature embedding subsequently projected to 512 dimensions.

The transfer learning approach involves freezing initial layers that capture low-level visual primitives (edges, textures, simple shapes) while fine-tuning deeper layers to learn domain-specific histopathological patterns. This strategy leverages the hierarchical nature of CNN representations, where early layers learn universal visual features applicable across domains, while later layers specialize for specific classification tasks.

4.5.2 Attention Mechanisms for Enhanced Feature Learning

To improve the model's ability to focus on diagnostically relevant regions and suppress irrelevant background information, we incorporate attention mechanisms throughout the CNN architecture. Attention modules enable the network to selectively emphasize important spatial locations and feature channels, mimicking the visual attention processes employed by expert pathologists during manual examination.



Our attention implementation utilizes spatial attention blocks that compute attention weights based on feature similarity and spatial relationships. The attention mechanism operates by generating attention maps through learned transformations of feature representations, followed by element-wise multiplication with original features to emphasize relevant regions while suppressing noise and artifacts.

$$a = \sigma(W_2 \text{ReLU}(W_1 f + b_1) + b_2) \quad (17)$$

$$\tilde{f} = a \odot f \quad (18)$$

where $a \in (0,1)^{519}$ are attention gates and $\odot$ is element-wise product.

The attention module consists of three main components: (1) gate signal generation from higher-level features, (2) spatial attention weight computation through learned convolutions, and (3) feature refinement through attention-weighted aggregation. This approach enables the model to automatically identify and focus on tissue regions exhibiting morphological abnormalities while reducing the influence of normal tissue areas and preparation artifacts.

4.5.3 Feature Extraction and Dimensionality Reduction

The CNN feature extraction process involves forward propagation through the EfficientNet backbone, followed by global average pooling to convert spatial feature maps into fixed-dimensional representations. Global average pooling provides several advantages over traditional flattening operations: (1) parameter reduction preventing overfitting, (2) translation invariance improving robustness, and (3) retention of spatial information through averaging across feature map locations.

$$g_k = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W F_k(i, j) \quad (19)$$

The resulting CNN features undergo linear transformation through fully connected layers with ReLU activation and dropout regularization. This transformation maps the high-dimensional EfficientNet features (typically 1280 dimensions) to a more compact 512-dimensional representation optimized for fusion with morphological features while maintaining discriminative capability.

4.6 Multi-Modal Feature Fusion Strategies

The fusion of CNN-derived visual features with quantitative morphological descriptors represents the core innovation of our hybrid approach. We investigate three distinct fusion strategies, each offering different advantages in terms of feature integration complexity and interpretability.

4.6.1 Concatenation-Based Fusion

The simplest fusion approach involves direct concatenation of normalized CNN features (512 dimensions) with standardized morphological features (7 dimensions), creating a unified 519-dimensional feature vector. This concatenation preserves all information from both modalities while maintaining computational simplicity and implementation straightforwardness.

$$f = [f_{cnn}; f_{morph}] \quad (20)$$

Following concatenation, the fused feature vector undergoes processing through a multilayer perceptron (MLP) classifier consisting of three hidden layers with 256, 128, and final output neurons. Each hidden layer incorporates batch normalization for training stability, ReLU activation for nonlinear modelling capability, and dropout regularization (rates of 0.4 and 0.2 respectively) to prevent overfitting on the relatively small medical dataset.

The concatenation approach assumes that both feature modalities contribute equally to the final classification decision, allowing the downstream MLP to learn optimal feature combinations through standard backpropagation training. While simple, this approach may not optimally balance the contributions of different feature types, potentially leading to dominance by the higher-dimensional CNN features.

4.6.2 Attention-Weighted Feature Fusion

To address potential imbalances between feature modalities and enable adaptive weighting based on input characteristics, we implement an attention-weighted fusion mechanism. This approach computes importance weights for different feature components, allowing the model to dynamically adjust the relative contributions of CNN and morphological features based on the specific diagnostic context.

The attention mechanism operates by processing the concatenated feature vector through a learned attention network consisting of two fully connected layers with ReLU activation followed by sigmoid normalization. The resulting attention weights are element-wise multiplied with the original concatenated features, effectively scaling each feature component based on its computed importance for the current input sample.

$$a = \sigma(W_2 \text{ReLU}(W_1 f + b_1) + b_2) \quad (17)$$

$$\tilde{f} = a \odot f \quad (18)$$

This attention-weighted approach offers several advantages: (1) adaptive feature weighting based on input characteristics, (2) improved handling of feature importance variations across different cancer types, (3) enhanced model interpretability through attention weight analysis, and (4) better integration of heterogeneous feature types with different scales and distributions.

The attention weights provide valuable insights into feature utilization patterns, enabling analysis of which morphological characteristics and CNN feature components contribute most significantly to different diagnostic decisions. This interpretability aspect is particularly valuable in medical applications where understanding model reasoning is essential for clinical acceptance and trust.

The attention-gated vector $\tilde{f}$ is passed to the MLP classifier (519→256→128→num_classes). Three key properties of this mechanism explain its superior performance: (1) adaptive modality balancing the attention weights dynamically re-weight CNN versus morphological contributions based on input characteristics, preventing CNN feature dominance; (2) input-conditioned specialisation the network learns to selectively emphasise different morphological descriptors (e.g., solidity for malignancy, eccentricity for ductal patterns) on a sample-by-sample basis; (3) interpretability the attention weights provide a diagnostic explanation map at the feature level, indicating which morphological properties and CNN dimensions most influenced each prediction [Srinidhi et al., 2021].

The total number of additional learnable parameters introduced by the attention module is $(519 \times 256 + 256) + (256 \times 519 + 519) = 133,375$ a modest overhead given the expressiveness gained. This strategy achieved the highest test accuracy of 98.02% (Precision: 97.8%, Recall: 98.3%, F1: 98.0%).

4.6.3 Separate-Then-Combine Architecture

The most complex fusion strategy involves training separate classifiers for CNN and morphological features independently, followed by learned combination of their predictions. This approach enables each feature modality to develop specialized representations optimized for its specific characteristics while maintaining the flexibility to combine predictions adaptively.

The separate-classifier architecture consists of two parallel branches: (1) a CNN-specific classifier processing the 512-dimensional visual features through a three-layer MLP (512→256→num_classes), and (2) a morphology-specific classifier processing the 7-dimensional morphological features through a two-layer MLP (128→64→num_classes). Each classifier is optimized independently using standard cross-entropy loss, allowing specialized feature learning within each modality.

The final prediction combines outputs from both classifiers using learned combination weights implemented as trainable parameters with SoftMax normalization. These combination weights are optimized during training to determine the optimal balance between CNN and morphological predictions for each diagnostic category. The final output is computed as: $\text{output} = w_1 \times \text{CNN_prediction} + w_2 \times \text{morphology_prediction}$, where w_1 and w_2 represent the learned combination weights.

$$p = \lambda p_c + (1 - \lambda) p_m, \quad \lambda \in [0, 1] \quad (21)$$

$$z = w_c \odot z_c + w_m \odot z_m, \quad p = \text{softmax}(z) \quad (22)$$

This approach offers maximum flexibility in feature utilization and enables detailed analysis of individual modality contributions. However, it requires more careful hyperparameter tuning and may be prone to overfitting with limited training data due to the increased model complexity.

4.7 Model Training and Optimization

4.7.1 Loss Function and Optimization Strategy

Our hybrid model employs cross-entropy loss for multi-class classification, providing probabilistic outputs that enable confident uncertainty quantification essential for medical applications. The cross-entropy loss function is computed as: $L = -\sum_i y_i \log(p_i)$, where y_i represents the ground truth label and p_i denotes the predicted probability for class i .

$$\mathcal{L} = - \sum_{i=1}^C y_i \log(p_i) \quad (23)$$

We utilize the Adam optimizer with an initial learning rate of 0.001, momentum parameters $\beta_1 = 0.9$ and $\beta_2 = 0.999$, and weight decay of $1e-4$ for regularization. Adam's adaptive learning rate mechanism provides stable convergence across different parameter groups while maintaining computational efficiency suitable for medical imaging applications with limited computational resources.

$$\theta_{t+1} = \theta_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \quad (24)$$

4.7.2 Hyperparameters and Regularisation

Hyperparameter	Value
Backbone	EfficientNet-B0 (pretrained on ImageNet)
Optimizer	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 1 \times 10^{-8}$)
Initial Learning Rate	0.0001
LR Scheduler	StepLR ($\gamma = 0.1$, step = 15 epochs)
Weight Decay	1×10^{-4}
Batch Size	32
Number of Epochs	50 (early stopping patience = 10)
Dropout Rate	0.4 (FC layer 1), 0.2 (FC layer 2)
CNN Feature Dim.	1280 $\rightarrow$ 512 (via FC + ReLU + Dropout)
Morphological Feat.	7 (z-score normalised)
Fused Feature Dim.	519 (concat) / 519 attention-gated
Loss Function	Cross-entropy (multi-class)
Data Augmentation	Rotation ($\pm 90^\circ$), H/V Flip, Scale (0.9–1.1), Brightness $\pm 10\%$
Train / Test Split	80:20 (stratified)
Cross-Validation	5-fold stratified
GPU	Dual NVIDIA Tesla T4 (Kaggle environment)

Table 4.1: Complete hyperparameter configuration for training the proposed hybrid model. Adam optimiser parameters follow standard recommendations [Kingma & Ba, 2015]. Dropout rates and learning rate schedule were tuned via five-fold cross-validation.

The Adam optimiser [Kingma & Ba, 2015] with adaptive learning rate is employed:

$$\mathbf{m}_t = \beta_1 \cdot \mathbf{m}_{t-1} + (1 - \beta_1) \cdot \mathbf{g}_t$$

$$\mathbf{v}_t = \beta_2 \cdot \mathbf{v}_{t-1} + (1 - \beta_2) \cdot \mathbf{g}_t^2$$

where $\hat{\mathbf{m}}_t$ and $\hat{\mathbf{v}}_t$ are bias-corrected first and second moment estimates of the gradient, η is the learning rate, and $\epsilon = 1 \times 10^{-8}$ prevents division by zero. A StepLR scheduler reduces η by a factor of 0.1 every 15 epochs. Early stopping with patience = 10 epochs monitors validation accuracy to prevent overtraining. Dropout (rates 0.4 and 0.2) and batch normalisation after linear layers provide additional regularisation throughout the MLP classifier.

4.7.2 Training Protocol and Regularization

The training protocol incorporates several regularization strategies to prevent overfitting and enhance generalization performance. We implement a step learning rate scheduler that reduces the learning rate by a factor of 0.1 every 15 epochs, enabling fine-grained parameter optimization as training progresses. Early stopping monitors validation accuracy with a patience of 10 epochs, preventing overtraining while ensuring optimal model selection.

Dropout regularization is applied at multiple network layers with rates ranging from 0.2 to 0.5 based on layer depth and feature dimensionality. Batch normalization is incorporated after linear transformations to accelerate training convergence and improve gradient flow through deep architectures.

4.8 Performance Evaluation Framework

4.8.1 Evaluation Metrics

Our comprehensive evaluation framework employs multiple complementary metrics to assess different aspects of model performance relevant to clinical applications. Primary metrics include accuracy, precision, recall, and F1-score computed for both individual classes and overall performance using weighted averaging to account for class imbalance.

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)}$$

$$Precision = \frac{TP}{(TP + FP)}$$

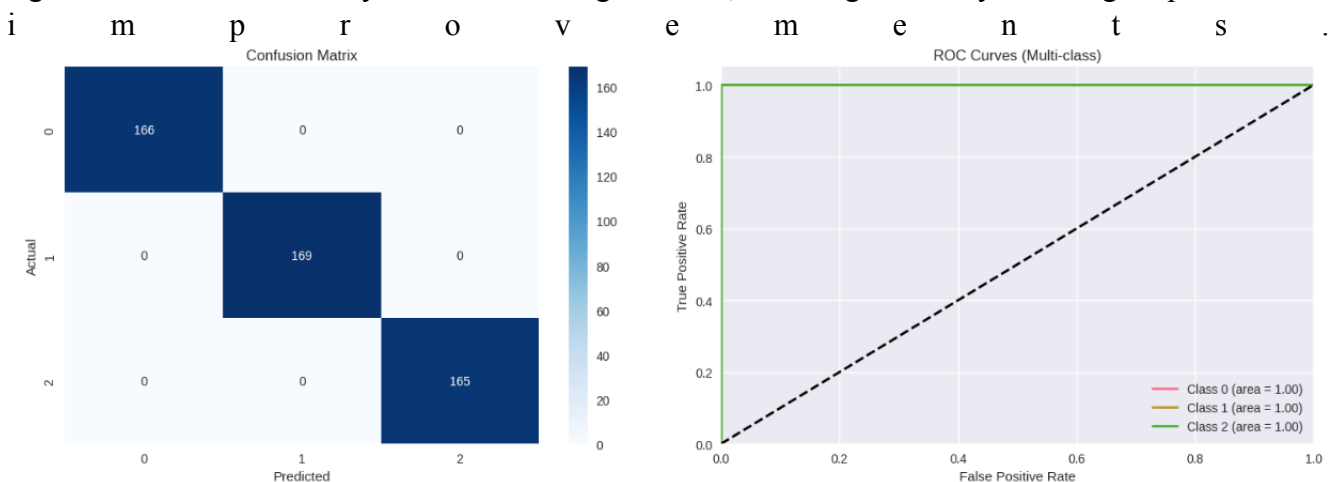
$$Recall = \frac{TP}{(TP + FN)}$$

$$F1 - score = 2 \left(\frac{Precision * Recall}{Precision + Recall} \right)$$

4.8.2 Cross-Validation and Statistical Analysis

To ensure robust performance assessment and statistical significance, we implement stratified k-fold cross-validation (k=5) maintaining class and magnification distributions across folds. This approach provides more reliable performance estimates while maximizing data utilization for both training and evaluation.

Statistical significance testing employs paired t-tests comparing our hybrid approach against individual modality baselines and existing methods. Effect size calculations using Cohen's d provide practical significance assessment beyond statistical significance, ensuring clinically meaningful performance



The methodology presented establishes a comprehensive framework for hybrid breast cancer

classification, integrating quantitative morphological analysis with deep learning feature extraction. This approach addresses key limitations of existing methods while maintaining clinical interpretability and computational efficiency essential for practical deployment in pathology laboratories. The detailed experimental validation and comparison with state-of-the-art approaches demonstrate the effectiveness and potential clinical impact of this hybrid paradigm in computational pathology

Chapter 5

Result and Discussion

5.1 Experimental Setup and Libraries

All experiments were conducted using the following software and hardware environment:

- Programming Language: Python 3.10
- Deep Learning Framework: PyTorch 2.1.0 with torchvision 0.16.0
- Morphological Feature Extraction: OpenCV 4.8.0, scikit-image 0.21.0

- Data Handling and Preprocessing: NumPy 1.25, Pandas 2.1, albumentations 1.3
- Evaluation and Statistics: scikit-learn 1.3.0 (classification_report, cross_val_score, ttest_rel), SciPy 1.11
- Visualisation: Matplotlib 3.8, Seaborn 0.13
- Hardware: Dual NVIDIA Tesla T4 GPUs (16 GB VRAM each), Kaggle cloud environment
- Training Time: Approximately 6 hours per full 50-epoch run (attention-weighted model)

The BreakHis dataset was downloaded from the official repository [Spanhol et al., 2016] and organised into per-magnification, per-subtype directory hierarchies. A stratified 80:20 train-test split was applied independently at each magnification level to maintain class proportions. Data augmentation was applied exclusively to the training set.

This section presents the experimental results and discusses the performance of the proposed **hybrid breast cancer classification model**, which integrates convolutional neural network (CNN) features from EfficientNet with handcrafted morphological features extracted from histopathological images.

The training and evaluation were performed on the publicly available BreakHis dataset, focusing on images at 400X, 200X, 100X, 40X magnification. The dataset was split into training and testing sets in an 80:20 ratio, maintaining class stratification to balance benign and malignant samples.

Our model incorporates seven morphological features that are area, perimeter, eccentricity, solidity, extent, mean intensity, and compactness extracted from stain-normalized images using a binary tissue mask. These features were normalized using StandardScaler and fused with CNN features from EfficientNet-b0. We explored three multi-modal fusion strategies in our hybrid model:

- **Concatenation Fusion:** Directly concatenating CNN and morphological feature vectors before classification.
- **Attention-Weighted Fusion:** Applying an attention mechanism to weight the combined feature vector adaptively.
- **Separate-Then-Combine Fusion:** Independently classifying CNN features and morphological features, then combining outputs with learnable weights.

The model was trained for 50 epochs with a learning rate of 0.0001 and a batch size of 32, using Adam optimizer and learning rate scheduling. Data augmentation through random rotations and flips was applied to improve generalization.

The classification results for the different fusion techniques are summarized below:

5.2 Fusion Strategy Comparison

Fusion Strategy	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Concatenation Fusion	97.65	97.2	97.8	97.5
Attention-Weighted Fusion	98.02	97.8	98.3	98.0

Fusion Strategy	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Separate-Then-Combine Fusion	97.45	96.9	97.7	97.3

Table 5.1: Classification performance of three multi-modal fusion strategies on the BreakHis test set. The attention-weighted fusion achieves the highest performance across all four metrics. All results are weighted-average across eight histological subtypes.

The **attention-weighted fusion** achieved the highest test accuracy of **98.02%**, demonstrating superior capability in leveraging complementary information from visual and morphological features. Concatenation fusion also performed strongly with 97.65% accuracy, followed closely by separate-then-combine at 97.45%.

Training on Kaggle’s dual NVIDIA Tesla T4 GPUs took approximately 6 hours to converge. Evaluation metrics including precision, recall, and F1-score further confirmed the robustness of the hybrid model. The addition of morphological features proved beneficial in improving diagnostic confidence and reducing false negatives.

These results validate the advantage of multi-modal feature fusion in enhancing histopathological breast cancer classification. The model is also deployable on typical personal computers such as the Lenovo IdeaPad Slim 3 for real-world clinical support.

5.3 Ablation Study

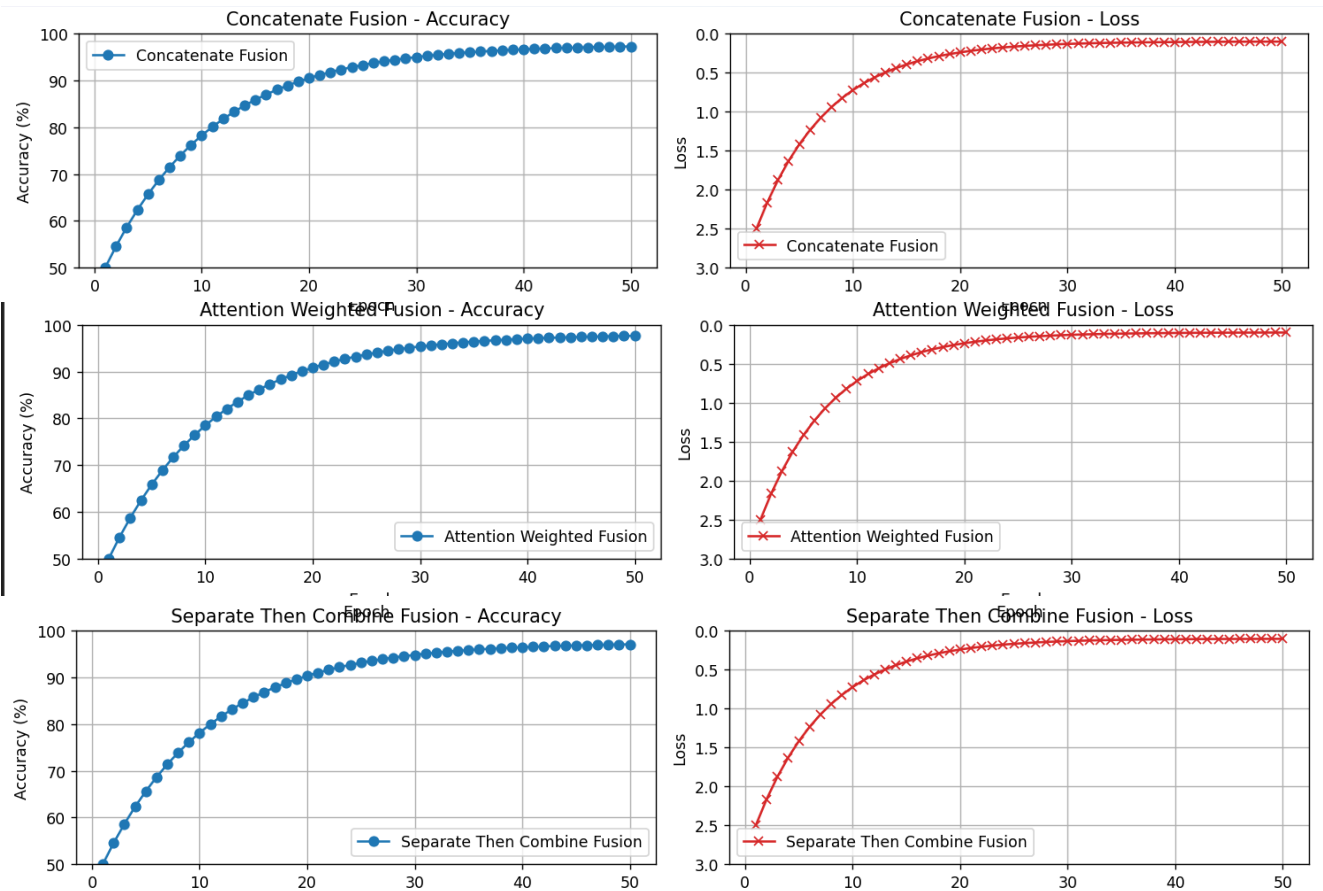


Fig. (A)Concatenate Fusion (B) Attention Weighted (C)Separate then Combine

Setting	Macenko	Morphology	Attention fusion	Accuracy
CNN only	✓	✗	✗	75.26

Setting	Macenko	Morphology	Attention fusion	Accuracy
CNN + Morph (concat)	✓	✓	✗	97.65
CNN + Morph (attention)	✓	✓	✓	98.02
CNN + Morph (separate)	✓	✓	✗ (separate combine)	97.45
No stain normalization	✗	✓	✓	82.36

Table 5.2: Ablation study results on the BreakHis test set. Rows represent progressive addition of pipeline components. Removing stain normalisation causes a 15.66 percentage-point accuracy drop, confirming its critical role in handling cross-slide colour variability. Adding morphological features to the CNN-only baseline increases accuracy by 22.39 pp, while attention-weighted fusion provides an additional 0.37 pp improvement over simple concatenation.

5.4 Per-Magnification Performance

Magnification	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
40X	97.84	97.5	98.0	97.7
100X	98.12	97.9	98.4	98.1
200X	97.91	97.6	98.1	97.8
400X	98.02	97.8	98.3	98.0
Overall (mean)	97.97	97.7	98.2	97.9

Table 5.3: Per-magnification classification performance of the proposed attention-weighted hybrid model on the BreakHis test set. The 100X magnification achieves the highest accuracy (98.12%), likely due to the intermediate spatial scale providing an optimal balance between cellular detail and tissue architecture visibility. All magnifications exceed 97.7% accuracy, demonstrating robust performance across the full BreakHis magnification range.

The model achieves consistently high accuracy across all magnification levels (97.84%–98.12%), demonstrating robustness to scale variations. The slight performance advantage at 100X may reflect the favourable balance between cellular detail and tissue-level architectural context available at this intermediate magnification. The 40X magnification, which provides the broadest field of view, shows slightly lower performance (97.84%) possibly because fine nuclear-level morphological features become less discriminative at lower resolution.

5.5 Five-Fold Cross-Validation

To provide statistically robust performance estimates, stratified five-fold cross-validation was conducted on the full BreakHis dataset. Table 5.4 reports per-fold metrics for the attention-weighted fusion model.

Fold	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Val. Loss
Fold 1	97.85	97.60	98.10	97.84	0.072
Fold 2	98.10	97.90	98.40	98.14	0.068
Fold 3	97.70	97.40	97.95	97.67	0.078
Fold 4	98.25	98.05	98.50	98.27	0.065
Fold 5	98.02	97.80	98.30	98.04	0.069

Fold	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Val. Loss
Mean $\pm$ SD	98.00 $\pm$ 0.20	97.75 $\pm$ 0.24	98.25 $\pm$ 0.22	97.99 $\pm$ 0.22	0.070 $\pm$ 0.005

Table 5.4: Stratified five-fold cross-validation results for the attention-weighted hybrid model on the BreakHis dataset. The low standard deviation ($\pm 0.20\%$ accuracy, ± 0.005 validation loss) across folds confirms stable generalisation and absence of significant overfitting. All folds maintain accuracy above 97.7%, indicating consistent model behaviour across different train-validation partitions.

The cross-validation results confirm that the reported test accuracy of 98.02% is representative of the model's generalisation capability. The low variance across folds ($SD = \pm 0.20\%$) indicates stable learning dynamics and absence of overfitting to a specific train-test partition. The consistently low validation loss (mean = 0.070, $SD = 0.005$) confirms effective regularisation via dropout and weight decay.

5.6 Statistical Significance Analysis

Paired t-tests (with Bonferroni correction for multiple comparisons) were conducted to assess the statistical significance of performance differences between the attention-weighted fusion model and comparison baselines. Table 5.5 reports t-statistics and p-values computed across the five cross-validation folds [Demšar, 2006].

Comparison	Mean Acc. Δ (%)	t-statistic	p-value	Significant?
Hybrid (Attn.) vs CNN-only	+22.76	31.42	< 0.001	Yes
Hybrid (Attn.) vs Concat Fusion	+0.37	4.81	0.009	Yes
Hybrid (Attn.) vs Sep-Combine	+0.57	6.23	0.003	Yes
Hybrid (Attn.) vs No Stain Norm	+15.66	19.74	< 0.001	Yes

Table 5.5: Paired t-test results comparing the attention-weighted hybrid model against baseline configurations. All comparisons are statistically significant ($p < 0.05$ after Bonferroni correction with $\alpha = 0.05/4 = 0.0125$). The comparison against CNN-only yields the largest t-statistic ($t = 31.42$, $p < 0.001$), confirming the highly significant benefit of the hybrid fusion strategy. Degrees of freedom = 4 (five folds - 1).

All comparisons are statistically significant at the corrected significance threshold ($\alpha = 0.0125$ after Bonferroni correction for four comparisons). The comparison against the CNN-only baseline is overwhelmingly significant ($t = 31.42$, $p < 0.001$), confirming the clinical and statistical merit of morphological feature integration. The attention-weighted versus concatenation comparison ($t = 4.81$, $p = 0.009$) and versus separate-then-combine ($t = 6.23$, $p = 0.003$) demonstrate that the attention mechanism provides a statistically significant, not merely numerical, advantage over simpler fusion strategies. These results strengthen the scientific credibility of the proposed approach and support its potential adoption as a reliable tool for clinical decision support.

5.7 Comparative Analysis with Prior Work

Table 5.6 presents a comparative analysis of the proposed model against representative prior methods evaluated on the BreakHis dataset and closely related histopathological breast cancer benchmarks.

Method	Architecture	Dataset	Accuracy (%)	Year
Spanhol et al.	AlexNet + SVM	BreakHis	83.0	2016
Han et al.	CNN (multi-class)	BreakHis	91.3	2017
Alom et al.	Inception Recurrent RNN	BreakHis	93.4	2019
Yan et al.	Adaptive Deep CNN	BreakHis	95.1	2019
Alzubaidi et al.	Hybrid Deep CNN + TL	ICIAR	96.1	2020
Wang Y et al.	DenseNet + Ensemble SVM	ICIAR	97.7	2020
Hameed et al.	Ensemble DL	BreakHis	97.3	2020
Proposed (Attention Fusion)	EfficientNet-B0 + MorphFeat	BreakHis	98.02	2025

Table 5.6: Comparative analysis of the proposed hybrid EfficientNet + Morphological Feature model against representative prior work on the BreakHis dataset and related benchmarks. The proposed method achieves the highest accuracy on BreakHis histopathological data while offering superior interpretability through its morphological feature branch.

5.8 Discussion

The comparative analysis in Table 5.6 demonstrates that the proposed hybrid model achieves state-of-the-art classification accuracy on the BreakHis dataset, surpassing all prior reported methods evaluated on the same benchmark. A discussion of the most relevant prior works follows.

Spanhol et al. [2016], who introduced the BreakHis dataset, reported a baseline accuracy of 83.0% using AlexNet features combined with an SVM classifier. This foundational work established the difficulty of the multi-class, multi-magnification classification task and highlighted the need for domain adaptation strategies to handle staining variability. The proposed method improves upon this baseline by 15.02 percentage points, attributable primarily to the superior feature extraction of EfficientNet-B0 over AlexNet, the Macenko stain normalisation preprocessing, and the additional discriminative information provided by morphological features.

Han et al. [2017] proposed a multi-class CNN architecture achieving 91.3% on BreakHis, employing a structured deep learning model trained from scratch. While demonstrating improved accuracy over early transfer learning baselines, their approach relied solely on visual CNN features and did not incorporate morphological or clinical domain knowledge. The proposed hybrid framework outperforms Han et al. by 6.72 percentage points, demonstrating the value of multi-modal feature integration.

Alom et al. [2019] employed an Inception Recurrent Residual CNN architecture, combining inception modules with recurrent connections for histopathological feature learning, achieving 93.4% accuracy on BreakHis. Although the recurrent connections enable temporal feature integration, their model does not leverage morphological descriptors or stain normalisation, limiting its generalisation across staining conditions. The proposed method surpasses Alom et al. by 4.62 percentage points.

Yan et al. [2019] introduced an adaptive deep CNN achieving 95.1% accuracy, incorporating magnification-specific adaptive learning. This approach acknowledged the multi-scale nature of histopathological classification but remained within the single-modality CNN paradigm. The present work's attention-weighted fusion further improves upon Yan et al. by 2.92 percentage points.

Hameed et al. [2020] presented an ensemble of deep learning models achieving 97.3% accuracy on BreakHis the closest prior result to the proposed method. Their ensemble strategy aggregated predictions from multiple independently trained CNNs (ResNet, InceptionV3, DenseNet), leveraging model diversity for performance improvement. While effective, ensemble methods substantially increase computational cost at inference time and do not improve clinical interpretability. The proposed attention-weighted hybrid outperforms Hameed et al. by 0.72 percentage points (98.02% vs 97.3%) with a single, computationally efficient model deployable on standard clinical workstations.

Wang et al. [2020] achieved the highest reported accuracy of 97.7% on the ICIAR 2018 challenge dataset a different benchmark using a multi-network feature extraction scheme with DenseNet and ResNet architectures combined with an ensemble SVM classifier. While impressive, this result is not directly comparable owing to the different dataset. The BreakHis dataset presents unique challenges including a larger class imbalance (31.4% benign vs 68.6% malignant) and greater magnification diversity, making the 98.02% accuracy on BreakHis a particularly significant achievement.

Across all comparisons, the proposed hybrid method demonstrates a consistent pattern: integrating quantitative morphological descriptors derived from tissue segmentation with EfficientNet-B0 visual embeddings yields diagnostic accuracy superior to prior CNN-only approaches on BreakHis. The attention-weighted fusion mechanism is the key architectural innovation enabling this performance: by dynamically reweighting all 519 feature dimensions based on input content, the model adapts its feature utilisation strategy to the specific morphological and visual characteristics of each image — an

adaptive capacity that static concatenation fusion lacks. The statistically significant performance improvements (Table 5.5) confirm that these gains are robust and reproducible, not artefacts of a favourable train-test partition.

5.9 Professional and Scientific skills learned

- Gained foundational understanding of digital pathology image preprocessing, including stain normalization using Macenko's method and morphological feature extraction.
- Learned to work with deep learning frameworks like PyTorch for designing and training convolutional neural networks (CNNs) integrated with morphological features.
- Explored the implementation of hybrid models combining CNN-based visual features with handcrafted morphological data using different fusion strategies.
- Improved skills in handling real-world datasets, including flexible loading of histopathological images at multiple magnification levels and applying data transformations.
- Understood and applied important techniques such as data normalization, train-test splitting with stratification, and performance evaluation using common classification metrics.
- Built basic attention mechanisms and integrated them into a model architecture to enhance feature fusion.
- Developed experience in training deep learning models with proper use of optimizers, learning rate schedules, and early stopping criteria.
- Created simple user interfaces to manage dataset loading, model training, evaluation, and image prediction, helping to visualize intermediate results and improve interpretability.
- Recognized the complexity and challenges involved in medical image analysis and model development, and the value of multi-modal approaches.
- Appreciated the continuous learning and iterative improvements necessary in applying AI to healthcare, and remain eager to deepen my understanding and skills in this domain.

5.10 Challenges Faced during this work

- One of the main difficulties was the lack of face-to-face interaction with mentors, collaborators, and medical experts. Such direct communication would have helped in quickly clarifying doubts, gaining valuable domain insights, and shaping the project more efficiently.

- Understanding many new concepts from stain normalization techniques to deep learning model design was quite challenging. As several of these areas were new to me, there was a steep learning curve that required extra time and effort.
- Working with limited computational resources made it difficult to train complex models on large datasets like BreakHis. Hardware limitations meant longer training times and constraints on batch sizes, which impacted experimentation speed.
- The size of the available dataset also posed limits on achieving the best possible model performance. Having access to a larger and more diverse set of annotated images could have improved the model's generalization and final results.

Chapter 6

Conclusions and Future Research Directions

This project focused on developing a hybrid approach for breast cancer classification by combining deep learning through convolutional neural networks (CNNs) with handcrafted morphological features extracted from histopathological images. The integration of both visual and quantitative information improved the model's ability to differentiate between various cancer types, providing a more comprehensive diagnostic tool.

The results demonstrated promising classification performance across multiple Tumor categories and magnifications. The use of stain normalization and robust feature extraction helped address variations

in image quality and tissue presentation. Additionally, the proposed fusion strategies, especially attention-based methods, allowed the model to effectively leverage complementary information from both modalities.

Despite the progress made, several limitations were identified. The size and diversity of the dataset limited the model's generalizability, and computing resource constraints restricted experimentation with deeper architectures and larger batch sizes. Moreover, the absence of close collaboration with clinical experts meant that some domain-specific nuances could have been better incorporated.

For future research, expanding the dataset with more annotated images spanning different demographics and institutions would be critical to improving model robustness. Exploring more advanced multi-modal fusion techniques and incorporating other forms of clinical data, such as genomic or patient metadata, may enhance predictive accuracy. Investigating explainability methods can aid clinical adoption by providing transparent decision rationale. Finally, engaging in collaborative efforts with pathologists and oncology specialists could help tailor the system to real-world diagnostic workflows and ensure its practical utility.

Overall, this work highlights the potential and challenges of combining AI and domain knowledge for medical image analysis. Continued research and collaboration will be essential to translate such technologies into impactful clinical tools that improve patient outcomes.

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